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Articles of footwear having therapeutic assemblies

Abstract

An article of footwear for warm-up and active recovery includes a sole and a light therapy system coupled to the sole. The light therapy system includes a plurality of LEDs, The article of footwear also includes a battery configured to power the light therapy system and control circuitry configured to control the light therapy system. The article of footwear may also include a vibration therapy system that includes a plurality of vibrating motors. The battery may power the vibration therapy system and the control circuitry may power the vibration therapy system.

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Background/Summary

RELATED APPLICATIONS (1) This application claims the benefit of U.S. provisional patent application 63/508,249, filed Jun. 14, 2023, titled “Articles of Footwear having a Vibration Assembly,” the entirety of the disclosure of which is hereby incorporated herein by this reference.

TECHNICAL FIELD

(1) This disclosure relates to articles of footwear, such as slides, that include one or more therapeutic assemblies, and more particularly to articles of footwear that provide targeted vibration and/or targeted light therapy to areas of a user's foot.

BACKGROUND

(2) Athletes and other individuals sometimes desire to use massage and therapeutic devices to warm up their muscles before working out and/or to recover after working out. Generic massage and therapeutic devices may not be ideal for a particular body part or muscle. Moreover, it can be time consuming to use a massage and therapeutic device for multiple body parts or muscles.

SUMMARY

(3) An article of footwear for warm-up, active recovery, and treatment of pain is provided according to some embodiments. The article of footwear may include a sole and a light therapy system coupled to the sole. In some embodiments, the light therapy system includes a plurality of LEDs. In some embodiments, the article of footwear includes a battery to power the light therapy system and control circuitry configured to control the light therapy system.

(4) In some embodiments, the article of footwear also includes a transparent layer disposed above the sole. In some embodiments, the plurality of LEDs is disposed between the transparent layer and the sole. In some embodiments, the article of footwear also includes a layer disposed above the sole. In some embodiments, the layer defines holes. In some embodiments, the article of footwear includes a transparent cover disposed in each hole. In some embodiments, each of the plurality of LEDs is aligned with one of the transparent covers. In some embodiments, the plurality of LEDs is arranged in a plurality of rows.

(5) In some embodiments, the article of footwear also includes a mounting layer disposed above the sole and a plurality of vibrating motors coupled to the mounting layer. In some embodiments, the battery is configured to power the plurality of vibrating motors and the control circuitry is configured to control the plurality of vibrating motors. In some embodiments, the mounting layer includes pockets. In some embodiments, each pocket is configured to receive a motor and each of the plurality of motors is disposed in one of the pockets. In some embodiments, the plurality of LEDs is disposed between the mounting layer and the sole. In some embodiments, the mounting layer defines a plurality of holes. In some embodiments, each hole is aligned with one of the plurality of LEDs. In some embodiments, a bottom surface of the mounting layer defines a cavity configured to receive the plurality of LEDs.

(6) In some embodiments, the article of footwear also includes a bracket disposed above the sole. In some embodiments, the bracket is configured to support the plurality of LEDs. In some embodiments, the bracket includes a longitudinal section and a plurality of transverse sections. In some embodiments, the battery is configured to be recharged with a wireless charging device.

(7) An article of footwear for warm-up and active recovery is provided according to some embodiments. The article of footwear may include a sole, a vibration system coupled to the sole, a light therapy system coupled to the sole, a battery to power the vibration system and the light therapy system, and control circuitry to control the vibration system and the light therapy system. In some embodiments, the vibration system includes a plurality of motors. In some embodiments, the light therapy system includes a plurality of LEDs.

(8) In some embodiments, the article of footwear also includes a mounting layer disposed above the sole. In some embodiments, the mounting layer has pockets, and each pocket can receive a motor. In some embodiments, each of the plurality of motors is disposed in one of the pockets and is configured to vibrate. In some embodiments, the mounting layer defines a plurality of holes. In some embodiments, each hole is aligned with one of the plurality of LEDs. In some embodiments, the plurality of LEDs is disposed between the mounting layer and the sole. In some embodiments, a bottom surface of the mounting layer defines a cavity configured to receive the plurality of LEDs. In some embodiments, the battery and the control circuitry are part of a control assembly and are disposed in a protective housing. In some embodiments, a bottom surface of the mounting layer defines a cavity configured to receive at least a portion of the control assembly.

(9) In some embodiments, the article of footwear also includes a bracket disposed above the sole and configured to support the plurality of LEDs. In some embodiments, the bracket includes a longitudinal section and a plurality of transverse sections. In some embodiments, the article of footwear also includes a plurality of transparent covers. In some embodiments, each transparent cover is disposed within one of the plurality of holes of the mounting layer. In some embodiments, the plurality of LEDs is disposed between the bracket and the plurality of transparent covers. In some embodiments, the battery is configured to be recharged with a wireless charging device. In some embodiments, the article of footwear is a slide. In some embodiments, the plurality of motors includes a first motor disposed in a forefoot region of the article of footwear. In some embodiments, the plurality of motors includes a second motor disposed in a midfoot region of the article of footwear. In some embodiments, the plurality of motors includes a third motor disposed between the midfoot region and a rearfoot region of the article of footwear.

(10) An article of footwear is provided according to some embodiments. The article of footwear may include a sole, a mounting layer coupled with the sole, a plurality of motors, a protective housing coupled to the mounting layer that contains a battery and control circuitry, and a switch coupled to the control circuitry and configured to turn the motors on and off. In some embodiments, the mounting layer has pockets with each pocket being configured to receive a motor. In some embodiments, each motor is disposed in one of the pockets and is configured to vibrate. In some embodiments, the battery is configured to power the motors and the control circuitry is configured to control the motors.

(11) In some embodiments, the article of footwear is a slide. In some embodiments, the article of footwear also includes one or more cables coupling the control circuitry and the plurality of motors. In some embodiments, the plurality of motors are eccentric rotating mass vibrating motors. In some embodiments, the plurality of motors includes five motors. In some embodiments, the plurality of motors includes at least one motor disposed in a forefoot region of the sole, at least two motors disposed in a midfoot region of the sole, and at least two motors disposed in a rearfoot region of the sole.

(12) In some embodiments, the pockets are disposed on a bottom of the mounting layer. In some embodiments, the pockets include five pockets. In some embodiments, the mounting layer is silicone. In some embodiments, the switch includes a button. In some embodiments, the button is disposed on an interior of the article of footwear. In some embodiments, the button is disposed such that a user's foot can operate the switch while wearing the article of footwear.

(13) In some embodiments, the switch is configured to transition between operating modes of the motors. In some embodiments, the operating modes includes different speeds of vibration. In some embodiments, the operating modes include different patterns of vibration. In some embodiments, the article of footwear also includes a charging port configured to receive a charger to charge the battery. In some embodiments, the protective housing is disposed in a foot arch section of the sole. In some embodiments, the battery is rechargeable. In some embodiments, the one or more cables includes three sets of cables. In some embodiments, the article of footwear also includes a strap coupled to the sole. In some embodiments, the strap extends from a lateral side of the sole to a medial side of the sole such that an area between the strap and the sole is configured to receive a wearer's foot. In some embodiments, the article of footwear is injection molded.

(14) In some embodiments, a top surface of the sole comprises cavities. In some embodiments, the cavities include a cavity for each of the plurality of motors. In some embodiments, the cavities include a cavity for the protective housing.

(15) An article of footwear for warm-up and active recovery is provided according to some embodiments. The article of footwear may include a sole and a vibration system coupled to the sole. In some embodiments, the vibration system includes a plurality of vibrating motors, a battery, control circuitry, and a switch. In some embodiments, the vibration system includes three vibration zones.

(16) In some embodiments, the three vibration zones include a front vibration zone having at least one vibrating motor of the plurality of vibrating motors disposed in a forefoot region of the sole. In some embodiments, the three vibration zones include a middle vibration zone having at least two vibrating motors of the plurality of vibrating motors disposed in a midfoot region of the sole. In some embodiments, the three vibration zones include a back vibration zone having at least two vibrating motors of the plurality of vibrating motors disposed in a rearfoot region of the sole. In some embodiments, the plurality of vibrating motors is at least partially embedded within the sole. In some embodiments, the article of footwear also includes a mounting layer coupled to the sole. In some embodiments, the mounting layer includes a plurality of pockets. In some embodiments, each of the plurality of vibrating motors is disposed in one of the plurality of pockets.

(17) A slide is provided according to some embodiments. The slide may include a sole, a strap coupled to the sole that extends from a lateral side of the sole to a medial side of the sole such that an area between the strap and the sole is configured to receive a wearer's foot, and a vibration system that is coupled to the sole. In some embodiments, the vibration system includes a plurality of vibrating motors, a battery, control circuitry, and a switch.

(18) The foregoing and other aspects, features, and advantages will be apparent from the DESCRIPTION and DRAWINGS, and from the CLAIMS if any are included.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Implementations will hereinafter be described in conjunction with the appended and/or included DRAWINGS.

(2) FIG. 1 shows a side perspective view of an article of footwear having a vibration assembly according to some embodiments.

(3) FIG. 2 shows a top schematic view of an article of footwear having a vibration assembly according to some embodiments.

(4) FIG. 3 shows a schematic of disassembled components of an article of footwear having a vibration assembly according to some embodiments.

(5) FIG. 4 shows a schematic of cabling for an article of footwear having a vibration assembly according to some embodiments.

(6) FIG. 5 shows a top view of an article of footwear (with the vibration assembly removed) according to some embodiments.

(7) FIGS. 6A-6C show potential areas of a user's foot that can be targeted for vibration therapy according to some embodiments.

(8) FIG. 7A shows a top view of an article of footwear having therapeutic assemblies according to some embodiments.

(9) FIG. 7B shows a top view of an article of footwear having therapeutic assemblies according to some embodiments.

(10) FIG. 7C shows a side view of an article of footwear having therapeutic assemblies according to some embodiments.

(11) FIG. 8 shows a cross-section schematic view of an article of footwear having therapeutic assemblies according to some embodiments.

(12) FIG. 9 shows a cross-section schematic view of an article of footwear having therapeutic assemblies according to some embodiments.

(13) FIG. 10 shows a cross-section schematic view of an article of footwear having therapeutic assemblies according to some embodiments.

(14) FIG. 11 shows a perspective schematic view of a mounting layer for an article of footwear having therapeutic

assemblies according to some embodiments.

(15) FIG. 12 shows a side schematic view of a mounting layer for an article of footwear having therapeutic assemblies according to some embodiments.

(16) FIG. 13A shows a top perspective view of an article of footwear having therapeutic assemblies according to some embodiments.

(17) FIG. 13B shows a side view of an article of footwear having therapeutic assemblies according to some embodiments.

(18) FIG. 14 shows a side view of an article of footwear having therapeutic assemblies according to some embodiments.

(19) FIG. 15 shows a top view of a sole for an article of footwear having therapeutic assemblies according to some embodiments.

(20) FIG. 16 shows a bottom perspective view of a mounting layer for an article of footwear having therapeutic assemblies according to some embodiments.

(21) FIG. 17 shows a bottom view of an article of footwear having therapeutic assemblies (with the sole removed) according to some embodiments.

(22) FIG. 18 shows a top perspective view of an article of footwear having therapeutic assemblies (with a mounting layer removed) according to some embodiments.

(23) FIG. 19 shows a top perspective view of an article of footwear having therapeutic assemblies (with a mounting layer and LED covers removed) according to some embodiments.

(24) FIG. 20 shows a cross section view of an article of footwear having therapeutic assemblies according to some embodiments.

(25) FIG. 21 shows a bottom perspective view of an article of footwear having therapeutic assemblies (with the sole and cover of a control assembly removed) according to some embodiments.

(26) FIG. 22A shows a charging system for an article of footwear having therapeutic assemblies according to some embodiments.

(27) FIG. 22B shows a charging system with an article of footwear having therapeutic assemblies according to some embodiments.

(28) FIG. 23 shows a bottom perspective view of an internal portion of an article of footwear having therapeutic assemblies according to some embodiments.

DETAILED DESCRIPTION

(29) Detailed aspects and applications of the disclosure are described below in the following drawings and detailed description of the technology. Unless specifically noted, it is intended that the words and phrases in the specification and the claims be given their plain, ordinary, and accustomed meaning to those of ordinary skill in the applicable arts.

(30) In the following description, and for the purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the various aspects of the disclosure. It will be understood, however, by those skilled in the relevant arts, that embodiments of the technology disclosed herein may be practiced without these specific details. It should be noted that there are many different and alternative configurations, devices, and technologies to which the disclosed technologies may be applied. The full scope of the technology disclosed herein is not limited to the examples that are described below.

(31) The singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a step” includes reference to one or more of such steps.

(32) The word “exemplary,” “example,” or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” or as an “example” is not necessarily to be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It is to be appreciated that a myriad of additional or alternate examples of varying scope could have been presented, but have been omitted for purposes of brevity.

(33) When a range of values is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. All ranges are inclusive and combinable. The term “plurality,” as used herein, means more than one.

(34) Throughout the description and claims of this specification, the words “comprise” and “contain” and variations of the words, for example “comprising” and “comprises,” mean “including but not limited to,” and are not intended to (and do not) exclude other components.

(35) As required, detailed embodiments of the present disclosure are included herein. It is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limits, but merely as a basis for teaching one skilled in the art to employ the present invention. The specific examples below will enable the disclosure to be better understood. However, they are given merely by way of guidance and do not imply any limitation.

(36) The present disclosure may be understood more readily by reference to the following detailed description taken in

connection with the accompanying figures and examples, which form a part of this disclosure. It is to be understood that this disclosure is not limited to the specific materials, devices, methods, applications, conditions, or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the claimed inventions.

(37) More specifically, this disclosure, its aspects and embodiments, are not limited to the specific material types, components, methods, or other examples disclosed herein. Many additional material types, components, methods, and procedures known in the art are contemplated for use with particular implementations from this disclosure.

Accordingly, for example, although particular implementations are disclosed, such implementations and implementing components may comprise any components, models, types, materials, versions, quantities, and/or the like as is known in the art for such systems and implementing components, consistent with the intended operation.

(38) The present disclosure relates to articles of footwear, such as slides, that include one or more therapeutic assemblies, and more particularly to articles of footwear that provide targeted vibration and/or targeted light therapy to areas of a user's foot. Athletes and other individuals sometimes desire to use massage and therapeutic devices to warm up their muscles before working out and/or to recover after working out. Generic massage and therapeutic devices may not be ideal for a particular body part or muscle. Moreover, it can be time consuming to use a massage and therapeutic device for multiple body parts or muscles. A massage and therapeutic device that is directed specifically to providing vibration therapy and/or light therapy to a user's foot is desirable, especially one that is incorporated into an article of footwear, such as slides.

(39) In some embodiments, an article of footwear comprises one or more therapeutic assemblies. The one or more therapeutic assemblies may include a vibration assembly, a light therapy assembly, or both. The article of footwear may have any of the features described in U.S. Provisional Application No. 63/508,249, filed Jun. 14, 2023, which is hereby incorporated by reference in its entirety into the present disclosure. For example, the vibration assembly may include any of the features described in U.S. Provisional Application No. 63/508,249, filed Jun. 14, 2023. In some embodiments, the one or more therapeutic assemblies may share components (e.g., power source, control system, etc.).

(40) In some embodiments, an article of footwear comprises a sole and a vibration assembly coupled to the sole. The vibration assembly may include one or more vibrating devices (e.g., vibrating motors), a battery, and control circuitry. The motors may be disposed in locations that facilitate targeted vibration to key areas of the foot, such as soft tissue structures, fascia, muscles, and regions of nerve innervation. In some embodiments, the article of footwear comprises a mounting layer for the motors, the battery, and/or the control circuitry. The mounting layer may be coupled with the sole. In some embodiments, the article of footwear comprises a switch. The switch may be configured to turn the vibrating motors on and off and/or to transition between different operating modes. In some embodiments, the article of footwear comprises a slide.

(41) In some embodiments, an article of footwear comprises a sole and a light therapy assembly coupled to the sole. The light therapy assembly may include one or more light-emitting diodes (LEDs), a battery, and control circuitry. The LEDs may be disposed in locations that facilitate targeted light therapy to key areas of the foot, such as soft tissue structures, fascia, muscles, and regions of nerve innervation. In some embodiments, the article of footwear comprises a mounting layer (such as an insole and/or other supporting structure) for the LEDs, the battery, and/or the control circuitry. The mounting layer may be coupled with the sole. In some embodiments, the article of footwear comprises an insole configured to allow light from the LEDs to reach a wearer's foot (e.g., a transparent insole, an insole with holes aligned with the LEDs). In some embodiments, the article of footwear comprises a switch. The switch may be configured to turn the LEDs on and off and/or to transition between different operating modes. In some embodiments, the article of footwear comprises a slide.

(42) In some embodiments, an article of footwear comprises both a vibration assembly (such as the one described above) and a light therapy assembly (such as the one described above). In some embodiments, the battery and the control circuitry may be used to control both therapeutic assemblies. In addition, the switch may be used to transition between various operating modes for both the vibration assembly and the light therapy assembly. In some embodiments, multiple switches may be used (e.g., one for each therapeutic assembly). Slides with vibration and/or light therapy and a built-in battery provide a portable, lightweight, and comfortable massage and therapeutic device that can provide targeted therapy to a wearer's feet. Such slides can be used for warm-up prior to working out and for active recovery after working out. Such slides can also be used for treatment of pain, inflammation, and joint stiffness.

(43) In some embodiments, the sole is shaped to provide massage, myofascial release, and/or a rolling benefit to the wearer. In addition, the sole's shape, material, or density may provide energy absorption, cushioning, and/or pressure distribution that leads to a more comfortable article of footwear (e.g., resulting in less stress on a wearer's joints). Thus, the sole's shape, material, or density may contribute to the therapeutic nature of the article of footwear (e.g., therapeutic effects related to pain, stiffness, and recovery).

(44) Embodiments of articles of footwear having one or more therapeutic assemblies are described below. Features that are discussed with respect to any embodiments below may also be used in conjunction with other embodiments, even if not expressly discussed with respect to other embodiments.

(45) An article of footwear **100** according to some embodiments is shown, for example, in FIG. 1. Article of footwear **100** may have a forefoot region **102**, a midfoot region **104**, and a rearfoot region **106**. Article of footwear **100** may

have a lateral side **101** and a medial side **103**. In some embodiments, article of footwear **100** comprises a sole **110**. Article of footwear **100** may be a slide, as shown in FIG. **1**. Thus, in addition to sole **110**, article of footwear **100** may comprise a strap **120** coupled to sole **110**. Strap **120** may be configured to extend from sole **110** at a lateral side **101** to sole **110** at a medial side **103** such that strap **120** and sole **110** together form an opening to receive a wearer's foot. In some embodiments, article of footwear **100** may be a different type of article of footwear, such as another type of sandal or a shoe with a sole and an upper section.

(46) In some embodiments, article of footwear **100** comprises a vibration assembly. As shown, for example, in FIGS. **2-3**, the vibration assembly may include one or more vibration devices **130** (e.g., motors **130**) and a control assembly **140**. The control assembly **140** may include a protective housing **150** containing a battery **160** and control circuitry **170** (see FIG. **3**). In some embodiments, article of footwear **100** comprises a mounting layer **112** to hold one or more of the motors **130** and any of the components of the control assembly **140**. These and other aspects of article of footwear **100** are discussed further below.

(47) In some embodiments, motors **130** are vibrating motors. For example, the motors **130** may be eccentric rotating mass (ERM) vibrating motors. Other types of vibrating motors may also be used. The vibration of the motors **130** provides massage therapy to targeted areas of the wearer's foot. Local vibration has been shown to have both prophylactic and reactive benefits related to circulation, recovery, and pain. Furthermore, it has been associated with being able to help different medical conditions related to both circulatory and neurological disease states. The vibration devices **130** can include vibration frequencies of between 0-300 Hz and vibration amplitudes of between 0.5-12 mm.

(48) In some embodiments, motors **130** are waterproof. In some embodiments, motors **130** are lightweight. For example, each motor **130** may be less than 10 grams. In some embodiments, each motor **130** may be less than 5 grams. Each motor **130** may be about 2 grams. In some embodiments, motors **130** may be designed to have minimum noise. In some embodiments, motors **130** may be pill-shaped motors. For example, motors **130** may have a diameter between 5 and 10 mm (e.g., 7 mm) and a length between 20 and 300 mm (e.g., 24.5 mm). In some embodiments, motors **130** with bigger dimensions may be used. Larger motors **130** may be beneficial in providing a stronger vibration. In some embodiments, motors **130** are coin-shaped instead of pill-shaped. Other shapes of motors **130** may also be used.

(49) The number and placement of motors **130** may vary in different embodiments. In some embodiments, article of footwear **100** may include one to ten motors **130**. For example, article of footwear **100** may include two, four, six, or eight motors. In some embodiments, article of footwear **100** includes more than ten motors **130**. Motors **130** may be positioned under or adjacent to various key areas of the foot, such as soft tissue structures, fascia, muscles, and regions of nerve innervation. Target areas of the foot may include one or more of plantar fascia, lateral plantar fascia, flexor digitorum brevis tendons, flexor hallucis longus tendon, adductor hallucis, lumbricales, flexor hallucis brevis, flexor digiti minimi brevis, plantar interossei, flexor digitorum brevis, abductor hallucis, abductor digiti minimi, plantar aponeurosis, quadratus plantae, flexor digitorum longus tendons, deep peroneal nerve, superficial peroneal nerve, tibial nerve, sural nerve, medial plantar nerve, lateral plantar nerve, medial calcaneal nerve, saphenous nerve, or other structures.

(50) As one example, FIG. **2** shows article of footwear **100** having five motors **130**. FIG. **2** is drawn schematically to show approximate locations of motors **130** (and the other components of the vibration assembly) with respect to a plan view of article of footwear **100**. Thus, these components would typically not be visible from a top view of article of footwear **100** but would instead be located underneath mounting layer **112** as discussed further below or otherwise embedded within sole **110**. Motors **130** may include a motor **130** disposed in the forefoot region **102**, two motors disposed in the midfoot region **104**, and two motors **130** disposed in the rearfoot region **106**.

(51) Motors **130** may include a motor **130** disposed in the forefoot region **102**, as shown in FIG. **2**. For example, a motor **130** may be disposed in the forefoot region **102** such that it would be disposed underneath the ball of a wearer's foot when article of footwear **100** is worn. In some embodiments, this forefoot motor **130** may be disposed centrally between lateral side **101** and medial side **103**. In some embodiments, this forefoot motor **130** may be disposed partially or entirely underneath strap **120**. For example, FIG. **2** shows the forefoot motor **130** located such that a portion of the forefoot motor **130** is underneath strap **120** and a portion of the forefoot motor **130** extends past strap **120** towards a front of article of footwear **100**. This forefoot motor **130** may be configured to provide massage therapy to the plantar fascia in the wearer's forefoot, superficial tendons and muscle in the forefoot (e.g., tendons of flexor digitorum brevis, flexor hallucis brevis), deep tendons and muscle in the forefoot (e.g., tendons of flexor digitorum longus, lumbricals, plantar interossei), and/or nerves in the forefoot (e.g., medial plantar nerve). Other locations of a forefoot motor **130** are also possible (e.g., to target a specific tendon, muscle, nerve, or other tissue structure). In some embodiments, multiple motors **130** may be disposed in forefoot region **102**.

(52) Motors **130** may include two motors **130** disposed in the midfoot region **104**, as shown in FIG. **2**. For example, a motor **130** may be disposed at a lateral side **101** of article of footwear **100** in the midfoot region **104** and another motor **130** may be disposed at a medial side **103** of article of footwear **100** in the midfoot region **104**. The medial side motor **130** in the midfoot region **104** may be disposed such that it would be disposed underneath and/or adjacent to the arch of a wearer's foot when article of footwear **100** is worn. Motors **130** in the midfoot region **104** may be aligned with each other such that their positions in a longitudinal direction are lined up or they may be offset from each other. In some embodiments, the midfoot motors **130** may be disposed partially or entirely underneath strap **120**. For example,

FIG. 2 shows the midfoot motors **130** located such that a portion of the midfoot motors **130** are underneath strap **120** and a portion of the midfoot motors **130** extends past strap **120** towards a back of article of footwear **100**. The medial midfoot motor **130** may be configured to provide massage therapy to the plantar fascia in the wearer's medial midfoot, superficial tendons and muscle in the medial midfoot (e.g., flexor hallucis brevis, abductor hallucis), deep tendons and muscle in the medial midfoot (e.g., tendons of tibialis posterior and flexor hallucis longus, flexor hallucis brevis), and/or nerves in the medial midfoot (e.g., medial plantar nerve, saphenous nerve). The lateral midfoot motor **130** may be configured to provide massage therapy to the plantar fascia in the wearer's lateral midfoot, superficial tendons and muscle in the lateral midfoot (e.g., flexor digiti minimi brevis, abductor digiti minimi), deep tendons and muscle in the lateral midfoot (e.g., tendons of fibularis brevis and fibularis longus), and/or nerves in the lateral midfoot (e.g., lateral plantar nerve, sural nerve). Other locations of midfoot motors **130** are also possible (e.g., to target a specific tendon, muscle, nerve, or other tissue structure). In some embodiments, a different number of motors **130** may be disposed in midfoot region **104** (such as only one motor **130** or more than two).

(53) Motors **130** may include two motors **130** disposed in the rearfoot region **106**, as shown in **FIG. 2**. For example, a motor **130** may be disposed at a lateral side **101** of article of footwear **100** in the rearfoot region **106** and another motor **130** may be disposed at a medial side **103** of article of footwear **100** in the rearfoot region **106**. The medial side motor **130** in the rearfoot region **106** may be disposed such that it would be disposed underneath and/or adjacent to the heel of a wearer's foot when article of footwear **100** is worn. Motors **130** in the rearfoot region **106** may be aligned with each other such that their positions in a longitudinal direction are lined up or they may be offset from each other. The medial rearfoot motor **130** may be configured to provide massage therapy to the plantar fascia in the wearer's medial rearfoot, superficial tendons and muscle in the medial rearfoot (e.g., flexor hallucis brevis, abductor hallucis), deep tendons and muscle in the medial rearfoot (e.g., tendons of tibialis posterior and flexor hallucis longus, flexor hallucis brevis, quadrates plantae), and/or nerves in the medial rearfoot (e.g., tibial nerve, saphenous nerve). The lateral rearfoot motor **130** may be configured to provide massage therapy to the plantar fascia in the wearer's lateral rearfoot, superficial tendons and muscle in the lateral rearfoot (e.g., flexor digiti minimi brevis, abductor digiti minimi, plantar aponeurosis), deep tendons and muscle in the lateral rearfoot (e.g., tendons of fibularis brevis and fibularis longus), and/or nerves in the lateral rearfoot (e.g., tibial nerve, sural nerve). Other locations of rearfoot motors **130** are also possible (e.g., to target a specific tendon, muscle, nerve, or other tissue structure). In some embodiments, a different number of motors **130** may be disposed in rearfoot region **106** (such as only one motor **130** or more than two).

(54) As already noted above, article of footwear **100** may have a different number of motors **130** and may use different locations for motors **130** other than those discussed above and shown in **FIG. 2**. In some embodiments, motors **130** may be placed to have different vibration zones. For example, the vibration assembly may provide two or more vibration zones. In some embodiments, the vibration zones may include a front vibration zone (e.g., a vibration zone provided by motor(s) **130** in forefoot region **102**), a middle vibration zone (e.g., a vibration zone provided by motor(s) **130** in midfoot region **104**), and a back vibration zone (e.g., a vibration zone provided by motor(s) **130** in rearfoot region **106**). In some embodiments, control assembly **140** may be configured to allow each vibration zone to be turned on individually or together with one or more of the other vibration zones.

(55) In some embodiments, motors **130** are coupled to mounting layer **112**, as shown, for example, in **FIG. 3**. Mounting layer **112** may include one or more pockets **114**. Each pocket **114** may be configured to receive a motor **130**. Thus, the number of pockets **114** on mounting layer **112** may be the same as the number of motors **130** discussed above. For example, mounting layer **112** may include five pockets **114**. In addition, the location of pockets **114** on mounting layer **112** and with respect to the rest of article of footwear **100** may be the same as the locations of motors **130** discussed above. In some embodiments, pockets **114** are disposed on a bottom of mounting layer **112**. Other components of the vibration assembly may also be coupled to mounting layer **112**. In some embodiments, the other components of the vibration assembly may be disposed underneath mounting layer **112** (e.g., embedded in a material of sole **110** or disposed in a cavity of sole **110** without any bonding connection to mounting layer **112**).

(56) Mounting layer **112** may be rigid enough to be durable but flexible to minimize noise of motors **130**. Various thicknesses of mounting layer **112** may be used (e.g., depending on the material of mounting layer **112**). Mounting layer **112** may be thin enough such that the vibration of motors **130** is felt by a wearer (so that the massage therapy is effective), while being thick enough to provide a reliable mounting structure for motors **130**. In some embodiments, mounting layer **112** is made of silicone. Other materials may also be used for mounting layer **112** (e.g., rubber, foam, etc.). In some embodiments, mounting layer **112** provides a soft material to be in contact with the wearer's foot (e.g., silicone, rubber, foam, etc.). In some embodiments, mounting layer **112** may be a plastic structure to support motors **130** with a soft material covering a top of mounting layer **112** to provide a soft surface to be in contact with the wearer's foot. In some embodiments, mounting layer **112** is coupled to sole **110**. The connection of mounting layer **112** to sole **110** may be any flexible, reliable, and waterproof connection. In some embodiments, mounting layer **112** is coupled to sole **110** using contact cement. In some embodiments, mounting layer **112** is glued to sole **110**. In some embodiments, mounting layer **112** is stitched to sole **110**. In some embodiments, mounting layer **112** is an additional layer that is added to sole **110**. For example, sole **110** may provide a hollow center surrounded by external surfaces forming sole **110**, and mounting layer **112** may be attached to sole **110** in the hollow center. In some embodiments, mounting layer **112** may be covered either from a top or bottom to enclose mounting layer **112** within sole **110**.

(57) In some embodiments, a top surface of mounting layer **112** forms a top surface of sole **110** on which a wearer's foot may rest. In some embodiments, a different material is disposed above mounting layer **112** to form a top surface of sole **110**. In some embodiments, article of footwear **100** does not include a mounting layer **112**. For example, the components of the vibration assembly (e.g., motors **130** and components of control assembly **140**, etc.) may be embedded within a material of sole **110** without using a mounting layer **112**.

(58) Control assembly **140** is configured to power and control motors **130** to provide massage therapy. In some embodiments, control assembly **140** may be disposed in an arch region **108** of article of footwear **100** (i.e., a region that corresponds to an arch of the wearer's foot when article of footwear **100** is worn), as shown in FIG. 2. Control assembly **140** may include protective housing **150** containing a battery **160** and control circuitry **170** (see FIG. 3). Control assembly **140** may also include cabling **180** coupling control circuitry **170** to motors **130**.

(59) In some embodiments, control circuitry **170** is configured to control operation of motors **130**. For example, control circuitry **170** may control the voltage provided to motors **130** to turn the motors on and off and/or to transition between different operating modes (vibration speeds, vibration patterns (e.g., vibration frequencies, intensities, or durations) vibration zones, etc.). In some embodiments, control circuitry **170** may also control the charging of battery **160**. In some embodiments, control circuitry **170** may comprise a printed circuit board. Control circuitry **170** may be contained within protective housing **150**.

(60) In some embodiments, a charging port **172** is coupled to control circuitry **170**. Charging port **172** may be configured to receive a charger to charge battery **160**. In some embodiments, charging port **172** is a USB-C port. Other types of ports may also be used for charging port **172**. Charging port **172** may be accessible from an exterior of article of footwear **100**. For example, charging port **172** may be disposed on an exterior wall of medial side **103** of sole **110**, such as at arch region **108** (see FIG. 2). In some embodiments, charging port **172** is supported and/or contained by protective housing **150**.

(61) In some embodiments, a switch **174** is coupled to control circuitry **170**. Switch **174** is configured to turn motors **130** on and off. For example, a wearer may use switch **174** to turn motors **130** on and off. In some embodiments, switch **174** is also configured to transition between operating modes of motors **130**. The operating modes may be different speeds of vibration, different patterns of vibration, or both. In some embodiments, the operating modes may include three different speeds. As one example, a first operation of the switch **174** may turn motors **130** on at a high speed, a second operation of switch **174** may change the motors to a medium speed, a third operation of switch **174** may change the motors to a low speed, and a fourth operation of switch **174** may turn the motors off. In addition, the operating modes may include different vibration zones. As one example, a user may first operate switch **174** to turn all motors **130** on, then to turn only a front vibration zone on (with the others off), then to turn only a middle vibration zone on (with the others off), then to turn only a back vibration zone on (with the others off), and then to turn all motors **130** off. Any other operating modes may also be used with the switch **174** allowing for a user to transition between any such operating modes. In some embodiments, multiple switches **174** may be included.

(62) In some embodiments, switch **174** may be a button **174** (e.g., a button that is pushed to operate). In some embodiments, button **174** is disposed on an interior of article of footwear. For example, button **174** may be disposed such that a user's foot can operate button **174** while wearing article of footwear **100**. Switch **174** may be located at medial side **103** in arch region **108** (see FIG. 2). Other locations may also be used. In some embodiments, switch **174** is supported and/or contained by protective housing **150**. In some embodiments, switch **174** may include multiple buttons **174** located in different areas of the article of footwear **100**.

(63) In some embodiments, battery **160** is configured to power motors **130**. In some embodiments, battery **160** is a rechargeable battery. For example, battery **160** may be a lithium-ion battery, such as a lithium-polymer battery. In some embodiments, battery **160** is configured to operate on a single charge for a run time of between one and six hours (e.g., two hours).

(64) In some embodiments, protective housing **150** contains battery **160** and control circuitry **170**. In some embodiments, protective housing **150** also contains charging port **172** and switch **174**. Protective housing **150** may be made of plastic, such as polycarbonate, acrylonitrile butadiene styrene, polyamide, or some other plastic material. Any other waterproof material may also be used. In some embodiments, protective housing **150** comprises a lower section **152** and an upper section **154**. Lower section **152** and upper section **154** may be joined together such that protective housing **150** is waterproof. For example, lower section **152** and upper section **154** may be joined using ultrasonic welding. In some embodiments, protective housing **150** may comprise connection points for cabling **180** to allow cabling **180** to electrically couple to control circuitry **170** inside protective housing **150**.

(65) In some embodiments, cabling **180** comprises one or more cables that couple the motors **130** to control circuitry **170** (and thus battery **160**). In some embodiments, cabling **180** is waterproof. Cabling **180** may be connected to motors **130** and to control circuitry **170** by being soldered.

(66) Cabling **180** may include three sets of cables, as shown, for example, in FIG. 4. The different sets of cables may correspond to vibration zones. Thus, a different number of sets of cables may also be used. In FIG. 4, the three sets of cables comprise a first set of two cables **182**, a second set of two cables **184**, and a third set of cables (having only one cable **186**). Each cable **182**, **184**, **186** extends from connection points **181** at protective housing **150** and/or control circuitry **170** to respective connection points at motors **130**. In some embodiments, the first set of cables **182** may

correspond to a back vibration zone and thus be connected to rearfoot motors **130** at connection points **183**. In some embodiments, the second set of cables **184** may correspond to a middle vibration zone and thus be connected to midfoot motors **130** at connection points **185**. In some embodiments, cable **186** may correspond to a front vibration zone and thus be connected to forefoot motor **130** at connection point **187**.

(67) Sole **110** may provide the main structure for article of footwear **100**. In some embodiments, the shape of sole **110** contributes to article of footwear **100** being a therapeutic device. For example, sole **110** may be shaped to provide massage to a wearer's foot while article of footwear **100** is being worn. Sole **110** may be shaped to facilitate myofascial release while a wearer is walking in article of footwear **100**. For example, sole **110** may provide a rolling benefit. In some embodiments, a shape of an external surface of mounting layer **112** (e.g., the surface facing the wearer's foot) may be shaped to provide massage, myofascial release, and/or rolling benefits. The shape of the external surface of mounting layer **112** may also be shaped to provide support for different areas of a wearer's foot.

(68) In some embodiments, sole **110** is configured to provide energy absorption, cushioning, and pressure distribution that increases the comfort for the wearer while using article of footwear **100**. For example, a shape, material, and/or density of sole **110** may be selected that lead to better energy absorption, cushioning, and pressure distribution, thus leading to a more comfortable experience. In some embodiments, for example, sole **110** may provide additional support under the wearer's foot arch.

(69) In some embodiments, sole **110** may be made of polyurethane or ethyl-vinyl acetate. In some embodiments, sole **110** may be made of a foam material. Strap **120** may be made of the same material as sole **110**. Sole **110** may be made using a variety of manufacturing methods. In some embodiments, sole **110** is injection molded. In some embodiments, strap **120** and sole **110** are molded with a single injection. In some embodiments, strap **120** is manufactured separately from sole **110** and then the two may be joined together. In some embodiments, sole **110** and/or strap **120** are 3D printed. In some embodiments, sole **110** (or the external-facing side of mounting layer **112**) may include a conductive material that is electrically grounded (e.g., by a wired connection to ground in control circuitry **170**) to provide a grounding or earthing effect to the wearer of the article of footwear **100** when the wearer's foot is in contact with the conductive material. In some embodiments, the grounding effect provided by the conductive material may provide therapeutic benefits to the wearer, such as reducing inflammation and pain and promoting wound healing. In some embodiments, the conductive material may be located in any region of article of footwear **100**, such as forefoot region **102**, a midfoot region **104**, and/or a rearfoot region **106**.

(70) In some embodiments, sole **110** may be configured to accommodate components of the vibration system. For example, sole **110** may comprise a surface **116** (as shown in FIG. 5) that defines a plurality of cavities sized and shaped to receive components of the vibration system. Surface **116** may be a top surface of the material that makes up sole **110**. Thus, the cavities may be formed during the injection molding process. In some embodiments, surface **116** defines motor cavities **117** configured to receive motors **130**. For example, motors **130** may be disposed within pockets **114** of mounting layer **112** and the pockets **114** and motors **130** may be fully or partially disposed within motor cavities **117**. Thus, the number of motor cavities in surface **116** of sole **110** may be the same as the number of motors **130** discussed above. For example, surface **116** may define five motor cavities **117**. In addition, the location of motor cavities **117** on surface **116** may be the same as the locations of motors **130** discussed above.

(71) In some embodiments, surface **116** defines protective housing cavity **118**. Protective housing cavity **118** may be disposed in arch region **108** of sole **110**. Protective housing cavity **118** is configured to receive protective housing **150**. In some embodiments, surface **116** defines a cavity **119** configured to receive charging port **172** and switch **174**. Cavity **119** may be disposed in arch region **108** of sole **110**. For example, cavity **119** may be disposed adjacent to protective housing cavity **118**.

(72) Many additional implementations are possible. For example, although FIGS. 1-3 and 5 show article of footwear **100** as a slide, other articles of footwear may also be used. In addition, although FIGS. 1-3 and 5 show particular number and locations of motors **130**, other variations may be used. For example, FIGS. 6A-6C illustrate other potential areas of a wearer's feet that may be targeted with vibrating motors for massage therapy using slides or other articles of footwear. FIG. 6A shows target areas **202** on a right foot **200** of a wearer and target areas **212** on a left foot of a wearer in a side view. FIG. 6B shows target areas **202** on a right foot **200** of a wearer and target areas **212** on a left foot of a wearer in a bottom or plantar view. FIG. 6C shows target areas **202** on a right foot **200** of a wearer and target areas **212** on a left foot of a wearer in a top or dorsal view.

(73) In some embodiments, an article of footwear may include a light therapy assembly in addition to or instead of a vibration assembly. For example, an article of footwear **300** according to some embodiments is shown in FIGS. 7A-7C having both a light therapy assembly and a vibration assembly. Features discussed with respect to article of footwear **100** above may also be implemented as part of article of footwear **300**. Similarly, features discussed with respect to article of footwear **300** below may also be implemented as part of article of footwear **100**. In addition, while article of footwear **300** comprises both a light therapy assembly and a vibration assembly, in some embodiments, an article of footwear comprises a light therapy assembly (such as the one discussed with respect to article of footwear **300**) and does not include a vibration assembly.

(74) Article of footwear **300** may have a forefoot region **302**, a midfoot region **304**, and a rearfoot region **306**. Article of footwear **300** may have a lateral side **301** and a medial side **303**. In some embodiments, article of footwear **300**

comprise a sole **310**. Article of footwear **300** may be a slide, as shown in FIGS. 7A-7C. Thus, in addition to sole **310**, article of footwear **300** may comprise a strap **320** coupled to sole **310**. In some embodiments, article of footwear **300** may include multiple straps **320** (e.g., two straps **320**). Strap(s) **320** may be configured to extend from sole **310** at lateral side **301** to sole **310** at medial side **303** such that strap(s) **320** and sole **310** together form an opening to receive a wearer's foot. In some embodiments, each strap **320** may comprise two parts, one extending from lateral side **301** of sole **310** and one extending from medial side **303** of sole **310**. The two parts may fasten together to form strap **320**. For example, the two parts may fasten with a buckle system (see FIGS. 7A-7B), a hook and loop system, a button system, a snap system, or other fastening systems. In some embodiments, article of footwear **300** may be a different type of article of footwear, such as another type of sandal or a shoe with a sole and an upper section. In some embodiments, the article of footwear **300** may include a fastening system and/or fastening mechanism that allows the footwear to be secured onto a wearer's foot. In some embodiments, the shoe's fastening mechanism may allow the footwear **300** to be adjusted or tightened properly to ensure a better fit for the wearer's foot, and to ensure an efficient delivery of the therapeutic treatment (e.g., light therapy and/or vibration therapy) to the wearer's foot.

(75) In some embodiments, article of footwear **300** comprises a vibration assembly, which may have the features discussed above with respect to article of footwear **100**, and a light therapy assembly. FIG. 7A shows article of footwear **300** with light therapy assembly turned off while FIG. 7B shows article of footwear **300** with light therapy assembly turned on. In some embodiments, a top layer **311** of article of footwear **300** is transparent. For example, top layer **311** may be made of a transparent material (e.g., transparent silicone). Configuring top layer **311** to be transparent facilitates light therapy for a wearer's foot from devices within article of footwear **300**, such as LEDs.

(76) Referring again to FIG. 7A, article of footwear **300** comprises a control assembly **340**. Control assembly **340** is disposed below top layer **311** (e.g., coupled to a mounting layer as discussed above). In some embodiments, control assembly **340** is visible through transparent top layer **311**, as shown in FIGS. 7A and 7B. Control assembly **340** is configured to power and control the vibration assembly and the light therapy assembly, and control assembly **340** may include any of the features discussed above for control assembly **140**. For example, control assembly **340** may be disposed in an arch region of article of footwear **300** (i.e., a region that corresponds to an arch of the wearer's foot when article of footwear **300** is worn).

(77) In some embodiments, article of footwear **300** comprises one or more vibration devices **330** (e.g., motors **330**). Motors **330** are disposed below top layer **311** (e.g., within a mounting layer as discussed above). In some embodiments, motors **330** are visible through transparent top layer **311**, as shown in FIGS. 7A and 7B. In some embodiments, motors **330** may have any of the features discussed above for motors **130**. In some embodiments, motors **330** have a diameter between 5 and 10 mm (e.g., 8 mm).

(78) The number and placement of motors **330** may vary in different embodiments. In some embodiments, article of footwear comprises three motors **330**. Motors **330** may include a first motor **330** in forefoot region **302** (which is shown just in front of forward strap **320** in FIGS. 7A and 7B), a second motor **330** in midfoot region **304** (which is partially shown just in front of the rearward strap **320** in FIGS. 7A and 7B), and a third motor **330** between midfoot region **304** and rearfoot region **306** (e.g., adjacent to and located rearward of control assembly **340**). Each of motors **330** may be disposed anywhere between medial side **303** and lateral side **301** of article of footwear **300** (e.g., centrally, closer to medial side **303**, or closer to lateral side **301**). Motors **330** may be positioned to provide targeted vibration therapy to certain portions of the wearer's foot (as discussed above).

(79) In some embodiments, article of footwear **300** comprises one or more light therapy devices **350** (e.g., light-emitting diodes (LEDs) **350**). In some embodiments, article of footwear comprises between two and thirty LEDs **350**, between ten and twenty LEDs **350**, or between twelve and sixteen LEDs **350** (e.g., 14 LEDs **350**). Different amounts of LEDs **350** (e.g., more than 30) may be used in some embodiments. The number and placement of LEDs **350** may vary in different embodiments.

(80) LEDs **350** may provide light therapy to help heal, increase circulation, stimulate metabolic processes, regenerate muscles, tendons, bones, joints, and other tissue, thus contributing to recovery after exercise and the treatment of both acute and chronic conditions associated with pain and inflammation. Red LEDs with visible red light in the spectrum of 620-700 nm and infrared LEDs producing near infrared (NIR) of wavelengths 700-2500 nm can be used to provide specific therapeutic fluence (dose in J/cm^{sup.2}). The therapeutic fluence is the product of a power density and treatment time that is both safe and effective in producing the physiological and neurological responses that produce the therapeutic benefit. In some embodiments, the power density of visible red light will be <300 mW/cm^{sup.2} with the power density of NIR <750 mW/cm^{sup.2}. In some embodiments, the fluence provided to the tissues will range between 20-1000 J/cm^{sup.2} depending on purpose of use and intended outcomes. Visible red light can be used for therapeutic benefits and as visual feedback to the user (e.g., wearer of the article of footwear **300**) to show that the device is working and producing light. LEDs **350** may be disposed in an array across article of footwear **300**. In some embodiments, LEDs **350** are arranged in a number of rows (e.g., one to six rows, two to four rows, six to ten rows, etc.), with each row having two to eight LEDs **350** (e.g., three to four LEDs). Each row of LEDs **350** may span across the width of article of footwear **300**. In some embodiments, LEDs **350** of each row are spaced equally apart from each other.

(81) In some embodiments, LEDs **350** are arranged in four rows. The foremost row (e.g., the row disposed in or closest

to forefoot region **302**) may be disposed below forward strap **320** (similar to what is shown in FIG. **13A**). The foremost row may include four LEDs **350**. In some embodiments, the foremost row of LEDs **350** is disposed at an angle relative to the transverse axis of article of footwear **300**. For example, LED **350** of the foremost row that is closest to lateral side **301** may be disposed rearwardly (i.e., closer to rearfoot portion **306**) of LED **350** of the foremost row that is closest to medial side **303**. In some embodiments, the foremost row is disposed rearwardly of motor **330** in forefoot region **302**.

(82) In some embodiments, a second row disposed adjacent to the foremost row comprises four LEDs **350**, as shown in FIG. **7B**. The second row of LEDs **350** may be disposed between two straps **320**. The second row of LEDs **350** may be disposed in forefoot region **302**, midfoot region **304**, or spanning across forefoot region **302** and midfoot region **304**. In some embodiments, no motors **330** are disposed between the foremost row and the second row of LEDs **350**. In some embodiments, the second row of LEDs **350** is disposed at an angle relative to the transverse axis of article of footwear **300**. For example, LED **350** of the second row that is closest to lateral side **301** may be disposed rearwardly (i.e., closer to rearfoot portion **306**) of LED **350** of the second row that is closest to medial side **303**. In some embodiments, the second row is disposed forwardly of motor **330** in midfoot region **304**.

(83) In some embodiments, a third row disposed adjacent to the second row comprises three LEDs **350**, as shown in FIG. **7B**. The third row of LEDs **350** may be disposed below or just behind a rearward strap **320**. The third row of LEDs **350** may be disposed in midfoot region **304**. In some embodiments, motor **330** in midfoot region is disposed between the second row and the third row of LEDs **350**. In some embodiments, the third row of LEDs **350** is disposed at an angle relative to the transverse axis of article of footwear **300**. For example, LED **350** of the third row that is closest to lateral side **301** may be disposed rearwardly (i.e., closer to rearfoot portion **306**) of LED **350** of the third row that is closest to medial side **303**. In some embodiments, the second row is disposed forwardly of motor **330** in midfoot region **304**.

(84) In some embodiments, a fourth (or rearmost) row disposed adjacent to the third row comprises three LEDs **350**, as shown in FIG. **7B**. The rearmost row of LEDs **350** may be disposed in midfoot region **304**, in rearfoot region **306** or spanning across midfoot region **306** and rearfoot region **306**. In some embodiments, motor **330** between midfoot region **304** and rearfoot region **306** region is disposed between the third row and the rearmost row of LEDs **350**. For example, motor **330** between midfoot region **304** and rearfoot region **306** may be disposed immediately forward of rearmost row of LEDs **350**. In some embodiments, the rearmost row of LEDs **350** is disposed at an angle relative to the transverse axis of article of footwear **300**. For example, LED **350** of the rearmost row that is closest to lateral side **301** may be disposed rearwardly (i.e., closer to rearfoot portion **306**) of LED **350** of the rearmost row that is closest to medial side **303**.

(85) Other arrangements of LEDs **350** may also be used. LEDs may be positioned to provide targeted light therapy to certain portions of the wearer's foot (similar to the vibration targets discussed above).

(86) In addition to or alternatively to the LEDs **350**, in some embodiments, article of footwear **300** may incorporate one or more materials for energy sources, such as graphene, carbon fiber, ceramics, or the like, to generate far infrared radiation (FIR) through heating for similar therapeutic benefits as discussed above. In some embodiments, materials such as graphene, carbon fiber, and/or ceramics may provide additional structural support to bear the weight of the wearer of the article of footwear **300**. In some embodiments, the support structure **455** may include one or more FIR-generating materials and/or heating elements for therapeutic purposes. Heat may provide a thermal-based therapeutic benefit to help with pain, joint stiffness and tension, reducing muscle spasms and muscle tone and improving blood flow. The aforementioned benefits fall under the umbrella of recovery from physical activity and the treatment of both acute and chronic conditions associated with pain and inflammation. Thermotherapy applied between the temperatures of 37 and 43 degrees Celsius can elicit the desired therapeutic effects while still being safe to users as they fall below the 44 degrees Celsius, which is the generally accepted maximum temperature before the user starts to experience adverse effects and pain.

(87) Materials producing far infrared radiation (FIR) in article of footwear **300** may provide therapeutic benefits to the user, including but not limited to reduced chronic muscle and joint pain and fatigue, wound healing, enhanced peripheral blood flow, and muscle tissue oxygenation. FIR is a band (50 to 1000 μm) in the infrared (IR) spectrum of electromagnetic radiation; the therapeutic fluence of FIR depends on the treatment duration as well as the two possible form factors of the device. The first being an electronically operated device which elicits greater FIR power densities in the tens of mW/cm^2 range by heating up a piece of graphene or ceramic material. The second form factor, utilizing textiles infused with germanium or similar materials, relies on energy produced by the body. This second form factor generates much lower power densities, being in the 0.1-5 mW/cm^2 range.

(88) In some embodiments, as shown, for example in FIG. **7C**, article of footwear comprises a charging port **372**. Charging port **372** is coupled to control assembly **340** and may be configured to receive a charger to charge a battery of control assembly **340**. In some embodiments, charging port **372** is a USB-C port. Other types of ports may also be used for charging port **372**. For example, charging port **372** may be a USB-A port, a DC jack connector, or other type of cable connector. Charging port **372** may be accessible from an exterior of article of footwear **300**. For example, charging port **372** may be disposed on an exterior wall of medial side **303** of sole **310**.

(89) In some embodiments, a switch **374** is coupled to control assembly **340** and configured to turn motors **330** and

LEDs 350 on and off. In some embodiments, switch 374 is disposed on an exterior wall of medial side 303 of sole 310. Similar to switch 174, switch 374 may be used to transition between operating modes in addition to turning therapeutic assemblies on and off. The operating modes may include any combination of characteristics (e.g., light therapy only, vibration therapy only, turning on all or a subset of motors 330, turning on all or a subset of LEDs 350, switching between different intensities, speeds, or patterns of vibration for motors 330, switching between different intensities of LEDs 350, a combination of the foregoing, etc.). In some embodiments, multiple switches 374 may be included. Switch 374 may have other characteristics of switch 174 described above. For example, switch 374 may be a button 374 (e.g., a button that is pushed to operate).

(90) Additional details of the inner arrangement of article of footwear 300 (e.g., motors 330 and LEDs 350) are illustrated with reference to FIGS. 8-10. FIG. 8 shows a cross section of article of footwear 300 (without including straps 320). Sole 310 provides a bottom layer of article of footwear 300 (e.g., a layer configured to contact the ground). Sole 310 is disposed beneath motors 330 and LEDs 350. In some embodiments, sole 310 is concave such that it receives other components of article of footwear 300 (e.g., control assembly 340, motors 330, LEDs 350, top layer 311, etc.). For example, sole 310 may include sidewalls 309 that extend from a base of sole 310 above other components of article of footwear 300. In some embodiments, sole 310 may be made of ethyl-vinyl acetate (EVA) or other material suitable for shoe/slide soles.

(91) The thickness of sole 310 may be different in various embodiments. A greater thickness of sole 310 may strengthen sole 310, particularly when a wearer is standing or putting weight on article of footwear 300. In some embodiments, a greater thickness helps facilitate vibration of motors 330, even when a wearer is standing. In some embodiments, as shown, for example, in FIG. 9, a thickness of sole 110 may be provided such that a distance between a bottom of motor 330 in forefoot region 302 and a base of sole 310 is between 12 mm and 20 mm (e.g., 15 mm). In some embodiments, a thickness of sole 110 may be provided such that a distance between a bottom of motor 330 in midfoot region 304 and a base of sole 310 is between 15 mm and 22 mm (e.g., 18 mm). In some embodiments, a thickness of sole 110 may be provided such that a distance between a bottom of motor 330 (disposed between midfoot region 304 and rearfoot region 306) and a base of sole 310 is between 15 mm and 22 mm (e.g., 19 mm). Other thicknesses may also be used.

(92) Referring again to FIG. 8, in some embodiments, article of footwear 300 comprises a mounting layer 312 disposed between top layer 311 and sole 310. In some embodiments, mounting layer 312 is made of EVA. Motors 330 may be coupled to mounting layer 312. For example, in some embodiments, mounting layer 312 includes pockets 314 for each motor 330, as shown in FIG. 8. In some embodiments, as shown in FIG. 9, pockets 314 include a spacer 315. Spacer 315 may be a protrusion of mounting layer 312 that extends below motor 330. In some embodiments, spacer 315 provides additional space for motor 330 to vibrate, even when article of footwear is being compressed (e.g., from the weight of a wearer).

(93) Referring again to FIG. 8, in some embodiments, mounting layer 312 includes holes 313 aligned with each LED 350, allowing light from LED 350 to reach (and shine through) transparent top layer 311. In some embodiments, each LED 350 may be disposed within hole 313 of mounting layer 312. In some embodiments, article of footwear 300 comprises a support structure 355 (such as a bracket 355) for LEDs 350. Bracket 355 may be configured to protect LEDs 350. In some embodiments, each row of LEDs 350 may have its own bracket 355. In some embodiments, each LED 350 may have its own bracket 355. In some embodiments, bracket 355 is one piece shaped to protect the entire array of LEDs 350.

(94) In some embodiments, bracket 355 may provide a base on which LED 350 may rest. In some embodiments, LEDs 350 rest on sole 310 and bracket 355 is disposed to surround LEDs 350 (e.g., around the sides of LEDs). In either of these situations, bracket 355 may extend around portions of LED 350, for example, to guide light from LED 350 up through hole 313 of mounting layer 312. In some embodiments, as shown, for example, in FIG. 10, bracket 355 comprises a hollow guide 356 that protrudes up through hole 313. In some embodiments, an internal portion of guide 356 may have a larger diameter towards the top of hole 313 than at a bottom of hole 313. This arrangement may provide an increased angle of light that can emanate from LED 350 through guide 356 (than if the inner diameter was constant), which may provide a higher intensity of LED light reaching a wearer, thus improving the light therapy.

(95) Bracket 355 may be coupled to mounting layer 312, sole 310, or both (e.g., with an adhesive). In some embodiments, bracket 355 is simply placed in location (e.g., in holes 313 of mounting layer 312 and/or above sole 310) without a direct coupling mechanism. In some embodiments, bracket 355 is made of a plastic material, such as polycarbonate. In some embodiments, bracket 355 is white to reduce the amount of light from LED 350 that is absorbed. Using a white bracket 355 may help improve the light therapy by increasing the intensity of LED light reaching a wearer.

(96) In some embodiments, control assembly 340 is disposed between mounting layer 312 and sole 310. Sole 310 and/or mounting layer 312 may include cavities or depressions to make space for control assembly 340. Control assembly 340 may be coupled to mounting layer 312, sole 310, or both. In some embodiments, control assembly 340 is simply placed in location (e.g., between mounting layer 312 sole 310) without a direct coupling mechanism.

(97) In some embodiments, rather than a transparent top layer 311 (e.g., made of silicone) and a separate mounting layer 312 (e.g., made of EVA), article of footwear 300 may include a single part made of polyurethane (e.g.,

thermoplastic polycarbonate (TPU)) and a transparent polycarbonate. For example, as shown in FIGS. 11 and 12, a single part **380** may be made that provides a soft layer **381** as a top surface. Layer **381** may be made of a TPU material and may be soft to provide for the wearer's comfort. In some embodiments, the same material that is used for layer **381** may be used to form pockets **314** (see FIG. 12).

(98) In some embodiments, single part **380** may be additionally made of a second material **382** that is transparent (e.g., transparent polycarbonate). Second material **382** may be disposed in locations within layer **381** as transparent covers **383** to fill holes in layer **381** that align with LEDs **350** (see FIG. 11). Thus, while layer **381** is not transparent, transparent covers **383** allow LED light to pass through to provide light therapy to a wearer's feet. In some embodiments, second material **382** may also be used to provide structure underneath layer **381** that provides structural support to protect LEDs **350**. In some embodiments, second material **382** is used to form spacers **315** to create support and empty space around motors **330** to facilitate their vibration (even when a wearer's weight is on article of footwear **300**) and also to reduce noise.

(99) An article of footwear **400** according to some embodiments is shown in FIGS. 13A-13B having both a light therapy assembly and a vibration assembly. Features discussed with respect to article of footwears **100** and/or **300** above may also be implemented as part of article of footwear **400**. Similarly, features discussed with respect to article of footwear **400** below may also be implemented as part of article of footwears **100** and/or **300**. In addition, while article of footwear **400** comprises both a light therapy assembly and a vibration assembly, in some embodiments, an article of footwear comprises a light therapy assembly (such as the one discussed with respect to article of footwear **400**) and does not include a vibration assembly.

(100) As shown in FIG. 13A, article of footwear **400** may have a sole **410** and one or more straps **420**. In some embodiments, article of footwear **400** comprises a vibration assembly, which may have the features discussed above with respect to article of footwear **100** or article of footwear **300**, and a light therapy assembly, which may have the features discussed above with respect to article of footwear **300**. In some embodiments, article of footwear comprises a mounting layer **412**. Mounting layer **412** may have some or all of the features of single part **380**, and some or all of the features of mounting layer **112**, or a combination of single part **380** and mounting layer **112**. For example, mounting layer **412** may comprise holes **413** (see FIG. 16) and transparent cover(s) **483** (see FIG. 13A) may be disposed in holes **413** at locations that are aligned with LEDs to allow for light therapy to be applied to a wearer's foot.

(101) In some embodiments, as shown in FIG. 13B, a switch **474** may be disposed on an exterior wall of sole **410**. Switch **474** may have the characteristics of switch **174** or **374**. Unlike article of footwear **300**, article of footwear **400** may use wireless charging (as discussed below) instead of having a charging port. In some embodiments, article of footwear **400** comprises a status light **475** disposed adjacent to switch **474**. Status light **475** may provide information to a user (e.g., low battery, fully charged, turned on, modes of operation, etc.) based on a color and/or a pattern of status light **475**. In some embodiments, as shown, for example, in FIG. 14, article of footwear **400** may include two switches, a switch **474** for the vibration assembly and a switch **476** for the light therapy assembly. Thus, switch **474** may be used to turn vibration motors on and off (and switch between various modes of operation), and switch **476** may be used to turn LEDs on and off (and switch between various modes of operation).

(102) Additional details of sole **410** are shown, for example, in FIG. 15. In some embodiments, sole **410** comprises a sidewall **409**. Sidewall **409** may be a raised edge around sole **410**. Sidewall **409** may extend above other components of article of footwear **400** (e.g., mounting layer **412**). In some embodiments, a top surface of sole **410** may define one or more cavities configured to receive other portions of article of footwear **400**. For example, top surface of sole **410** may define a cavity **417** for each motor in the vibration assembly of article of footwear **400**. In FIG. 15, sole **410** defines three cavities **417** for motors. In some embodiments, top surface of sole **410** defines a cavity **418** configured to receive all or a portion of a control assembly for the vibration assembly and light therapy assembly of article of footwear **400**. In some embodiments, top surface of sole **410** defines a cavity **419** for the LEDs (and associated structure) of the light therapy assembly of article of footwear **400**. In some embodiments, cavities **417**, **418**, and **419** may overlap or be nested within each other. For example, cavity **418** may be the deepest cavity and may partially overlap with one or more cavities **417** and with cavity **419**. Cavities **417** may be deeper than cavity **419** and may partially or entirely overlap with cavity **419**. In some embodiments, a portion of cavities **417** is configured to provide extra space to allow motors to vibrate, even when article of footwear **400** is being compressed.

(103) Additional details of mounting layer **412** are shown, for example, in FIG. 16. In some embodiments, mounting layer **412** defines holes **413** therethrough. Holes **413** may be positioned to align with LEDs of the light therapy assembly of article of footwear **400**. In some embodiments, mounting layer **412** comprises pockets **414** configured to hold motors of the vibration assembly of article of footwear **400**. Pockets **414** may be disposed on a bottom of mounting layer **412**. In FIG. 16, mounting layer **412** comprises three pockets **414** for motors. In some embodiments, a bottom surface of mounting layer **412** defines a cavity **415** configured to receive all or a portion of a control assembly for the vibration assembly and light therapy assembly of article of footwear **400**. In some embodiments, a bottom surface of mounting layer **412** defines a cavity **416** for the LEDs (and associated structure) of the light therapy assembly of article of footwear **400**.

(104) Additional details relating to the light therapy assembly is shown, for example, in FIGS. 17-19. FIG. 17 shows a bottom view of article of footwear **400** with sole **410** removed. Portions of mounting layer **412** are visible, including

pockets **414** with cabling of the motors extending out of the pockets **414**. Other portions of mounting layer **412** are covered by a control assembly **440** and support structure **455**. In some embodiments, control assembly **440** is disposed to be aligned with cavity **415** (see FIG. 16). Control assembly **440** may be operably connected with switch **474** and status light **475**.

(105) In some embodiments, support structure **455** comprises structure (e.g., bracket **455**) to contain LEDs. In some embodiments, bracket **455** comprises a longitudinal section **456** that extends longitudinally along a side of article of footwear **400**. For example, longitudinal section **456** may extend from a rearfoot portion of article of footwear **400** to a forefoot portion of article of footwear **400**. Branching from longitudinal section **456** is a series of transverse sections **457** extending transversely across article of footwear **400**. Each transverse section **457** may be associated with a row of LEDs. Thus, the number of transverse sections **457** may be equal to the number of rows of LEDs in article of footwear **400**. In some embodiments, a control assembly section **458** also extends transversely from longitudinal section **456** to control assembly **440**. An electrical connection from control assembly **440** to LEDs may be contained within bracket **455**. In some embodiments, an electrical connection from control assembly **440** to motors of article of footwear **400** may also be contained within bracket **455**. For example, the cabling extending from pockets **414** may extend into bracket **455**. In some embodiments, bracket **455** is U-shaped to be disposed below and to the sides of LEDs when article of footwear **400** is worn.

(106) FIG. 18 shows a top view of article of footwear **400** with mounting layer **412** removed. In some embodiments, transparent covers **483** (which are configured to fill holes **413** in mounting layer **412**) are disposed over bracket **455**. In some embodiments, transparent cover **483** is configured to mate with bracket **455** so as to enclose LEDs within bracket **455** and transparent cover **483**. In some embodiments, a row of transparent covers **483** may be one piece, as shown in FIG. 18. Each row of transparent covers **483** may correspond to a row of LEDs, a row of holes **413**, and a transverse section **457** of bracket **455**. In some embodiments, transparent covers **483** comprise lenses for LEDs.

(107) FIG. 19 shows a similar view as FIG. 18, but transparent covers **483** are also removed to reveal LEDs **450** and LED support **465**. LED support **465** may provide a physical and electrical connection between each LED **450** and control assembly **440**. In some embodiments, LED support **465** is shaped to fit within bracket **455**. LEDs **450** may rest on top of LED support **465**. In some embodiments, LED support **465** comprises a longitudinal section **466** that extends longitudinally along a side of article of footwear **400**. For example, longitudinal section **466** may extend from a rearfoot portion of article of footwear **400** to a forefoot portion of article of footwear **400**. Branching from longitudinal section **466** is a series of transverse sections **467** extending transversely across article of footwear **400**. Each transverse section **467** may be associated with a row of LEDs **450**. Thus, the number of transverse sections **467** may be equal to the number of rows of LEDs **450** in article of footwear **400**. In some embodiments, portions of longitudinal section **466** disposed between transverse sections **467** may be softer than other portions of LED support **465**. This may help make electrical connections between LEDs **450** more reliable, particularly while a wearer is walking in article of footwear **400**, and thus improve reliability of LEDs **450** themselves in providing light therapy.

(108) In some embodiments, a control assembly section **468** also extends transversely from longitudinal section **466** to control assembly **440**. LED support **465** may provide an electrical connection from control assembly **440** to LEDs **450**. In some embodiments, LED support **465** may also provide a portion of an electrical connection from control assembly **440** to motors **430** of article of footwear **400**. For example, cabling of motors **430** may electrically connect to LED support **465**.

(109) The cross-section view shown in FIG. 20 shows various components of the light therapy assembly of article of footwear **400** assembled together (including transparent cover **483**, mounting layer **412**, LED **450**, LED support **465**, bracket **455**, sole **410**, and control assembly **440**). Control assembly **440** may include a protective housing **444**, control circuitry **442** (e.g., a printed circuit board (PCB)), and a battery **460** (which may have the features discussed above for protective housing **150**, battery **160**, and control circuitry **170**, while also powering and controlling operation of light therapy assembly). In some embodiments, control circuitry **442** may include a wireless communication module (e.g., for Bluetooth connection, near-field communication (NFC), Wi-Fi connection, or the like). In some embodiments, each shoe in a pair of articles of footwear **400** may connect or pair with each other and work in tandem via a connection through the wireless communication module in control circuitry **442**. In some embodiments, the wireless communication module may allow for the article of footwear **400** to connect to and communicate with an external device (e.g., a mobile device, tablet, computing device, or the like), via Bluetooth, NFC, Wi-Fi, or other networks. In some embodiments, a wearer of the article of footwear **400** may control operation of the article of footwear **400** (e.g., powering on/off the footwear, powering on/off the vibration and/or light therapies, adjusting the levels of vibration and/or light therapies, etc.) through the external device that is connected to the article of footwear **400**. In some embodiments, the wearer of the article of footwear **400** may program or select a schedule of treatment for the vibration and/or light therapies in the footwear **400** using the interface of the connected external device. In some embodiments, the schedule of treatment may include a combination of time intervals and/or varying durations for treatment, such as for example, vibration and/or light therapy for a duration of 10 minutes every hour for 5 hours.

(110) In some embodiments, a wireless charging module **472** (as shown in FIGS. 20 and 21) is disposed underneath battery **460**. Wireless charging module **472** is configured to charge battery **460** when paired with a charging device **500** (see FIGS. 22A and 22B). In some embodiments, charging device **500** comprises a connector **510**, a cable **515**, a

flexible portion 520, and one or more charging pads 530. In some embodiments, connector 510 may be configured to plug into a power source (e.g., a power outlet, a USB port, etc.). In some embodiments, connector 510 comprises a USB connector. In some embodiments, cable 515 extends from connector 510 to flexible portion 520. Flexible portion 520 may be disposed between two charging pads 530, allowing for charging device 500 to fold into a more compact configuration (see FIG. 22A). Having two charging pads 530 allows for a pair of articles of footwear 400 to be charged simultaneously. Each article of footwear 400 is placed on a charging pad 530 such that charging pad 530 is aligned with charging module 472 is aligned with charging pad 530 (see FIG. 22B). Thus, article of footwear 400 may be charged wirelessly (instead of with a charging port). In some embodiments, wireless charging of article footwear 400 may allow for the device's electronics to be sealed and waterproof.

(111) Variations to article of footwear 400 are also possible. For example, instead of having a single LED support 465 with a longitudinal section 466, each transverse section 467 may be its own LED support, as shown, for example, in FIG. 23. In place of longitudinal section 466 and control assembly section 468, an electrical connection from control assembly 440 may be provided with cables 469 to connect each transverse section 467. This configuration may also make the electrical connections between LEDs 450 more reliable, particularly while a wearer is walking in article of footwear 400, and thus improve reliability of LEDs 450 themselves in providing light therapy.

(112) Further implementations are within the CLAIMS. It will be understood that implementations of articles of footwear having one or more therapeutic assemblies include but are not limited to the specific components disclosed herein, as virtually any components consistent with the intended operation of various articles of footwear having one or more therapeutic assemblies may be utilized. Accordingly, for example, it should be understood that, while the drawings and accompanying text show and describe particular implementations of articles of footwear having one or more therapeutic assemblies, any such implementation may comprise any shape, size, style, type, model, version, class, grade, measurement, concentration, material, weight, quantity, and/or the like consistent with the intended operation of articles of footwear having one or more therapeutic assemblies.

(113) The concepts disclosed herein are not limited to the specific articles of footwear having one or more therapeutic assemblies shown herein. For example, it is specifically contemplated that the components included in particular articles of footwear having one or more therapeutic assemblies may be formed of any of many different types of materials or combinations that can readily be formed into shaped objects and that are consistent with the intended operation of the articles of footwear having one or more therapeutic assemblies. For example, the components may be formed of: rubbers (synthetic and/or natural) and/or other like materials; glasses (such as fiberglass), carbon-fiber, aramid-fiber, any combination therefore, and/or other like materials; elastomers and/or other like materials; polymers such as thermoplastics (such as ABS, fluoropolymers, polyacetal, polyamide, polycarbonate, polyethylene, polysulfone, and/or the like, thermosets (such as epoxy, phenolic resin, polyimide, polyurethane, and/or the like), and/or other like materials; plastics and/or other like materials; composites and/or other like materials; metals, such as zinc, magnesium, titanium, copper, iron, steel, carbon steel, alloy steel, tool steel, stainless steel, spring steel, aluminum, and/or other like materials; and/or any combination of the foregoing.

(114) Furthermore, articles of footwear having one or more therapeutic assemblies may be manufactured separately and then assembled together, or any or all of the components may be manufactured simultaneously and integrally joined with one another. Manufacture of these components separately or simultaneously, as understood by those of ordinary skill in the art, may involve 3-D printing, extrusion, pultrusion, vacuum forming, injection molding, blow molding, resin transfer molding, casting, forging, cold rolling, milling, drilling, reaming, turning, grinding, stamping, cutting, bending, welding, soldering, hardening, riveting, punching, plating, and/or the like. If any of the components are manufactured separately, they may then be coupled or removably coupled with one another in any manner, such as with adhesive, a weld, a fastener, any combination thereof, and/or the like for example, depending on, among other considerations, the particular material(s) forming the components.

(115) In places where the description above refers to particular implementations of articles of footwear having one or more therapeutic assemblies, it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these implementations may be applied to other implementations disclosed or undisclosed. The presently disclosed articles of footwear having one or more therapeutic assemblies are, therefore, to be considered in all respects as illustrative and not restrictive.

Claims

1. An article of footwear for warm-up and active recovery, the article of footwear comprising: a sole; a light therapy system coupled to the sole, the light therapy system comprising a plurality of LEDs; a bracket disposed above the sole and configured to support the plurality of LEDs; a mounting layer disposed above the sole, wherein the plurality of LEDs is disposed between the mounting layer and the sole, and wherein the mounting layer defines a plurality of holes, each hole aligned with one of the plurality of LEDs; a plurality of vibrating motors coupled to the mounting layer; a battery configured to power the light therapy system and the plurality of vibrating motors; and control circuitry configured to control the light therapy system and the plurality of vibrating motors.

2. The article of footwear of claim 1, further comprising a transparent layer disposed above the sole, wherein the plurality of LEDs is disposed between the transparent layer and the sole.
 3. The article of footwear of claim 1, further comprising: a transparent cover disposed in each hole, wherein each of the plurality of LEDs is aligned with one of the transparent covers.
 4. The article of footwear of claim 1, wherein the plurality of LEDs is arranged in a plurality of rows.
 5. The article of footwear of claim 1, wherein the mounting layer comprises pockets, each pocket configured to receive a motor, and wherein each of the plurality of motors is disposed in one of the pockets.
 6. The article of footwear of claim 1, wherein a bottom surface of the mounting layer defines a cavity configured to receive the plurality of LEDs.
 7. The article of footwear of claim 1, wherein the bracket comprises a longitudinal section and a plurality of transverse sections.
 8. The article of footwear of claim 1, wherein the battery is configured to be recharged with a wireless charging device.
 9. The article of footwear of claim 1, further comprising one or more materials configured to generate at least one of far-infrared radiation (FIR) and heat.
 10. An article of footwear for warm-up and active recovery, the article of footwear comprising: a sole; a vibration system coupled to the sole, the vibration system comprising a plurality of motors; a mounting layer disposed above the sole, the mounting layer having pockets, each pocket configured to receive a motor, wherein each of the plurality of motors is disposed in one of the pockets and configured to vibrate; a light therapy system coupled to the sole, the light therapy system comprising a plurality of LEDs, wherein the plurality of LEDs is disposed between the mounting layer and the sole, and wherein the mounting layer defines a plurality of holes, each hole aligned with one of the plurality of LEDs; a battery configured to power the vibration system and the light therapy system; and control circuitry configured to control the vibration system and the light therapy system.
 11. The article of footwear of claim 10, wherein a bottom surface of the mounting layer defines a cavity configured to receive the plurality of LEDs.
 12. The article of footwear of claim 10, wherein the battery and the control circuitry are part of a control assembly and are disposed in a protective housing.
 13. The article of footwear of claim 12, wherein a bottom surface of the mounting layer defines a cavity configured to receive at least a portion of the control assembly.
 14. The article of footwear of claim 10, further comprising a bracket disposed above the sole and configured to support the plurality of LEDs, wherein the bracket comprises a longitudinal section and a plurality of transverse sections.
 15. The article of footwear of claim 14, further comprising a plurality of transparent covers, each transparent cover disposed within one of the plurality of holes of the mounting layer, wherein the plurality of LEDs is disposed between the bracket and the plurality of transparent covers.
 16. The article of footwear of claim 10, wherein the battery is configured to be recharged with a wireless charging device.
 17. The article of footwear of claim 10, wherein the article of footwear comprises a slide.
 18. The article of footwear of claim 10, wherein the plurality of motors comprises a first motor disposed in a forefoot region of the article of footwear, a second motor disposed in a midfoot region of the article of footwear, and a third motor disposed between the midfoot region and a rearfoot region of the article of footwear.
 19. An article of footwear for warm-up and active recovery, the article of footwear comprising: a sole; a light therapy system coupled to the sole, the light therapy system comprising a plurality of LEDs; a bracket disposed above the sole and configured to support the plurality of LEDs; a mounting layer disposed above the sole, wherein the mounting layer comprises a plurality of holes and a transparent cover disposed in each hole, wherein each one of the plurality of LEDs is aligned with the transparent cover disposed in each hole; a plurality of vibrating motors coupled to the mounting layer; a battery configured to power the light therapy system and the plurality of vibrating motors; and control circuitry configured to control the light therapy system and the plurality of vibrating motors.
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